

1-What is Von Neumann Architecture?

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|---------------------------------|--------------------------------|-----|
| 1)Single Bus, Unified Memory | 2) Single Bus, Separate Memory | |
| 3) Multiple Bus, Unified Memory | 4) Single Bus, Separate Memory | 5)? |

2-What is Harvard Architecture?

- | | | |
|---------------------------------|--------------------------------|-----|
| 1)Single Bus, Unified Memory | 2) Single Bus, Separate Memory | |
| 3) Multiple Bus, Unified Memory | 4) Single Bus, Separate Memory | 5)? |

3-Is there any better architecture?

- 1)Yes 2)No

4-What is the main memory?

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|---------------------------|----------------------|-----|
| 1)A memory out of the CPU | 2)An internal Memory | |
| 3)A high speed disk | 4)A high performance | 5)? |

5-What is "Internal cache"?

- | | | |
|---|------------------------------|-----|
| 1)A high speed memory | 2)Content addressable Memory | |
| 3)A high speed memory with same technology of the CPU | 4)2 and 3 | 5)? |

6-What is "External cache"?

- | | | |
|---|------------------------------|-----|
| 1)A high speed memory | 2)Content addressable Memory | |
| 3)A high speed memory with same technology of the CPU | 4)2 and 3 | 5)? |

7-What is a register?

- | | | |
|---|------------------------------|-----|
| 1)A high speed memory | 2)Content addressable Memory | |
| 3)A high speed memory with the same technology of the CPU | 4)2 and 3 | 5)? |

8- Between "External cache" and "Internal cache" which one is faster?

- | | | |
|-------------------------|------------------|------------------------------|
| 1)External Cache | 2)Internal Cache | 3)It is technology dependent |
| 4)The speed is the same | 5)? | |

9- Between "Internal cache" and "Register" which one is faster?

- | | |
|------------------|------------|
| 1)Internal Cache | 2)Register |
|------------------|------------|

10- A minimum configuration ALU can be made using:

- 1)an Adder and NAND gate 2) 1)an Adder and Or Gate

11-A More sophisticated ALU can be made using:

- 1)an Adder and NAND gate 2) 1)an Adder and Or Gate 3)?

12-Do you think that “cache level3” is useful?

- 1)Yes 2)No

13- Why we use level1 and level2 caches?

APPROVED

By_____ **Date**_____